Inês Hipólito - Generative models of the mind: neural connections and cognitive integration

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um hi hello everyone I'm quite excited to be here and first I'd like just to thank you organizers for putting together such a great conference thanks so much for the invite I'd like to start by saying that I'm a philosopher okay so bear with me I don't have a lot of formal stuff that you like this is also brand new research so I'm testing the waters here and see if there's something that I should carry on doing or not so let me know all the feedback is very welcome okay so having said that let me start by just um just saying that what I want what I want to research here is a the impact of action and I take a subjective experience to be in the action okay so the idea here is to look at our action is what may or may not allow explaining the neural connectivity in the brain okay so the presentation will be the five of in two parts our one metaphysical the other one is neuroanatomy so first what I'm going to do is I'm going to look at the fundamental distinction between the metaphysics of inference between functional and effective connectivity this will then allow me to focus on effective connectivity and I want to look at particularly on the oculomotor system as sort of a study case and I'm going to look at that from the active inference point of view okay so um yet we all know are from the last three days at least that the integration of the brain areas is something that is quite difficult to assess and one way to do that is by looking to the in terms of the brain connectivity and our cognition integrates from that point of view which is what motivates me to look at the metaphysics of the connectivity so first I want to just focus on the motivation ok so the idea here is that there is this oppression operational distinction between functional and effective connectivity and this is important because it allows us to determine the nature of the inferences that can be made about cognitive integration and also most importantly the questions there one should be asking and I'm following here Tristan in doing so so what I want to do is just look at the differentiation on the one hand on the causal modeling in effective connectivity and on the other hand as it is opposed to the procedures used to characterize functional connectivity okay so let me start by looking at functional connectivity just the very basics the aim is that it is inferred on the basis of temporal correlations between time series of different brain regions in a sort of measurements of this modular neural activity these connections are can then be in the stood in terms of these causal symmetry that is in terms of these probabilities and temporal order okay so the aim here and what is more important is that the aim is like that to identify which are the segregated brain regions that are most likely to connect in order to carry on a specialized kind of function or for a task to main and I think Han and colleagues are captured is really well when they say that the functional specialization and anatomical segregation of neural networks in the brain imply that it is organized as distributed hierarchical network of highly specialized networks of spiking neurons so the idea is that there's this time-resolved functional connectivity which is claimed as crucial to displaying a sort of a balance between on the one hand functional segregation and on the other hand integration of brain networks and here's where modularity networks comes handy and particularly important because it attempts to do just starts to find the patterns of correlations between functionally segregated areas on the one hand it's significant integration on the other hand and that's where it becomes quite important on the other hand we have effective connectivity and here we look at the causal dependency is more like sort of what motivates the coupling or the direct influence between for example a and B this will rest obviously is basically on a model of that influence that is called a generative model this kind of models the generative models are then compared as the probabilistic strategy is at the heart of these analysis in effective connectivity so following frist and it tells us that effective connectivity explains brains networks by virtue of accounting for the dynamic activity activity dependent and the dependence on a model of these interactions one example of these studies on effective connectivity can be found for example in Sokolove and colleagues when they look at visual sensitivity to biological motion that cause changes in the connectivity and actually how this active interaction relates to behavior so this analysis of effective connectivity it can be said that it is both neural and activity lab and it is on this account that these models are conducted or carried by effective connectivity are said to be quite dynamic so the idea is that artists if they do not suppose that there are there is this invariant structure that can be captured anatomically okay so um just let me let us just have a look at how we can just a systematize these differences connectivity which will allows that allows allows me to then move forward to effective connectivity okay on the one hand we have effect of the from the functional connectivity as being a sort of identifying temporal correlations or covariances on the other hand effective connectivity as a matter of explaining these causal dependences we have also on functional connectivity are the aim to map from imaging data which is it's physiological consequences to its diagnostic class which is a categorical kind of cause on the other hand we have an effective connectivity sort of a model comparison as with just saw generative models of coupling among these hidden brain states sorry on the other hand and effective connectivity we have mapping from causes to consequences so the other way around so we have own functional connectivity a sort of a predictive model whereas on effective connectivity it's more of a model comparison Jarett of model of coupling among hidden brain states ok good so some people think and come forward to say that well effective connectivity is more powerful and less ambiguous ok and this would this would be the idea also followed by free stem 2011 well it that when it tells us well this is the case because it captures both the topological pairwise connections as well as direct causal influences and we can see that in these graphs in which on the first one the structural causal modeling we can look at the topological pairwise connections and on the second one with the dynamic causal model modelling we can see that but we can also see the direct causal influence so this is what motivates me to now focus on the affective connectivity and now we can explain that by looking at the oculomotor system ok so what makes the oculomotor system such an interesting case is that it is a distributed network composed by the cerebral cortex basal ganglia and the cerebellum and the idea here is that neural messages from these regions that combined to generate signals to the extraocular musk muscles to move the eyes so to show this kind of coupling between perception or vision and action I will and the causal dependencies on this neural connection I will first look at the neuroanatomy and then I'm gonna look at the active inference okay so just a very basics I just want to flesh out just the saccadic eye so we have the dynamic integration of vision in action why well because the action takes the form of the saccadic eye movements that is a series of these discrete fixations and there are interposed by rapid and rapid movements so while we may select a new target for fixation it is necessary to apply forces that decelerate the eyes towards their target so this is the very basics of it now moving at how we could explain this relation between perception and action through active inference we can say that well by forming hypotheses this saccades can be deployed as experiments to adjudicate among alternatives so unpacking this a little bit objects in the visual field become hypotheses or explanations that take the form of something like what would I see if I looked there so Bayesian filtering equations can be used and implemented to planning as ever as inference which can generate both saccadic and smooth pursuit eye movements so importantly the associated message passing will map pretty well on to the known connectivity in neuroanatomy of the brainstem which obviously didn't end don't have time to explain here okay so this is some the main point is just that there are these signals that are carried by each eye to the brainstem and they can be classified in two broad categories and this is what what the main point that I want to stress out here that's the X T receptive and the proprioceptive so on the acceptive as we all know we have the visual signals to the optic nerve and then on the proprioceptive level there is this extra ocular muscles into the oculomotor nerves which control the angular man Angela position and velocity of each eye okay so in other terms visual signals are then generated through an identity mapping from the position of the eyes right so this means that what the eye sees what I see depends entirely on where they look this becomes quite important because then proprioceptive states are what comes to ensure that these mapping from the hidden States to two outcomes and the result of these outcomes is that then is the reception so vision is depending or in a dependence in a probabilistic sense on proprioception which is to say in action so the idea is that vision is dependent on action and action will be what explains these visual connections okay um this causality between action and perception can be explained in terms of the free energy principle which is that living systems must minimize the original free energy over time and this minimization of the free energy through action and perception is referred to as active inference active inference and very briefly is just this theory there is a front derives from the principle dispersion of this the state in order to continue to exist in a sort of a meaningful way and in order to do that to find the things that are relevant to their own existence so that they don't die they draw these predictions about the sensations that they will encounter and actively sample them in them to reduce these uncertainty or mathematically the expected Bayesian surprise okay um now this becomes a form of what Thomas paw and quarry stone called the active vision and this is the bit that I'm very interested on on working on and building from so this is the idea that this activision is a charity of process these filled circles are what gives rise to the sensory data and then we have on the other hand the charity of model which is the unfilled circles which proposes this data is generated now an arrow connecting two variables indicates that the second variable is conditionally dependent on the first and is what I'm particularly interested because this this means that eyes are conditionally dependent on target fixation prediction okay so this fictive locations then will cause changes in position in the generative model and which brings us to conclude that then action is part of these Jarett of process okay so then how do we put all of this together well these equations can be implemented by passing messages between populations of neurons sending messages here are excitatory prediction errors while descending messages are the inventory predictions and it is this patterns that characterizes what we know is a predictive coding now we see on the right hat on the left hand side here we got equations that describe these kind of gradient descent on variational free-energy neural message passing and on the on the right side we have these equations mapping to the scheme of this generative model so what happens is that we have a map of the influences between each population with excitatory and inhibitory connections okay good now bringing these to an end and take a message what I want to say and what I want to research on building on this is that well first I think there are few quite relevant and important points to take from here is not accept receptive vision and proprioception emerge from a single imperative which is to contextualise each other in the sense that they all are reducing minimizing the free energy ineffective conic connectivity terms then what motivates the specific establishment of certain synaptic connections and visions is the active exploration and this is the bit that I'm quite interested of the world by focusing on what appear as salient to a member of a certain phenotype now in other times this means that active inference on active domain the active dimension of vision is what explains these visual connections so which means now we're not just simply collecting or picking out what we see in the world there is an active dimension to it now finally the effective connectivity captures these essential what is essential to the construction and explanation of the our article process in the visual world okay now coming to the last slide as to future work I just want to say that it would be quite important and crucial I think to find these links between network topology in neural dynamics that can address the empirical problem of how the brain adapts in a dynamic and flexible way and according to the demands of the cognitive interaction with the environment which I take to be the subjective experience now this was not a talk about subjective experience this doesn't mean that I don't think that there is such thing as subjective experience I think there is it's inaction and by the way I wrote a paper on what it feels like to prove a mathematical equation so but actually these top was more about the role that subjective experience plays in neural connections or and also neural deployment so the question that I think is quite relevant to to work on future research is how this subjective experience acts on the ways that the brain connects and reorganizes to connect in cognitive terms so this is me thank you very much for your attention [Applause]